

HUANJUN WANG

SENIOR SOFTWARE ENGINEER | BUILD SYSTEMS, CI/CD & DEVELOPER INFRASTRUCTURE

Bellevue, WA · Huanjun.wang@proton.me · Open to relocation
U.S. Permanent Resident (Green Card Holder) · No Sponsorship Required

SUMMARY

Senior Software Engineer with over 10 years across production software and developer infrastructure. Has taken **three separate build-platform onboarding and modernization programs** from planning through cutover, twice as tech lead, on **Microsoft CloudBuild** — a distributed build service with **remote execution and content-addressed caching** — across polyglot monorepos ranging from single-digit gigabytes to tens of gigabytes and serving 200 to 2,000+ engineers. That work spans **dependency-graph analysis, affected-target incremental builds**, deterministic and binary-equivalent output validation to keep cross-machine cache reuse correct, and build observability through dashboards, metrics, and logs. Works **full-stack** alongside it — frontend, APIs, backend services, and databases — with production engineering experience on large distributed fleets and petabyte-scale data pipelines under regulated workloads.

TECHNICAL SKILLS

Build & Developer Infrastructure: Microsoft CloudBuild (distributed remote execution, content-addressed build cache, build telemetry), MSBuild, NMake, dependency-graph analysis, affected-target incremental builds, deterministic and binary-equivalent output validation, build observability (dashboards, metrics, logs), CI/CD integration, release engineering, Bazel

Languages: Python, TypeScript, C++, C#

Cloud & Delivery: Azure, GCP, GitHub Actions, Terraform, Docker

Leadership: Technical design, project execution, mentoring, cross-team platform integration, rollout planning, incident remediation

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Microsoft, eDiscovery

Redmond, WA | 2022 – Present

Build, CI & Deployment Modernization

- Led the modernization of build, CI, and deployment workflows for a polyglot eDiscovery repository spanning C#, C++, Scala, and TypeScript and serving roughly 200 engineers; owned technical direction and led a two-engineer vendor team.
- Onboarded the repository onto CloudBuild distributed execution and build caching, then tuned it alongside repository-wide dependency mapping and engine and framework upgrades, cutting typical build times from about an hour to a few minutes.
- Built dashboards tracking build duration and cache state, and used CloudBuild logs to diagnose intermittent pipeline failures across distributed build machines.

Large-Scale Production Systems

- Contributed directly to production systems that included a 10,000+ VM Elasticsearch fleet and migration pipelines handling roughly 100 PB of data; implemented recovery, data validation, observability, tenant-isolation, and RBAC capabilities for regulated workloads.

Full-Stack Product Engineering

- Delivered eDiscovery features end to end as the primary focus of the role, across frontend workflows, APIs, backend services, and databases, partnering with feature teams and the central Office engineering team throughout.
- Implemented collection throttling and job-level deduplication that reduced failures by two orders of magnitude and eliminated redundant downstream work.

Tech Lead / Software Engineer II | Microsoft, SharePoint Search

Suzhou, China | 2017 – 2021

Build Platform Migration

- Served as tech lead for an independent CloudBuild onboarding across the SharePoint Search codebase, used by roughly 200 engineers; owned the technical plan and cutover sequencing while mentoring two junior engineers through the rollout.
- Cut typical incremental build times from hours to minutes by combining distributed remote execution and build caching with dependency-graph-driven affected-target selection, validating build correctness against the pre-migration baseline.
- Coordinated with the central build-platform team to keep daily developer workflows running throughout the transition.

Software Engineer | Microsoft, SharePoint Engineering Velocity

Suzhou, China | 2015 – 2017

Build Infrastructure Modernization

- Part of the team that onboarded SharePoint's tens-of-gigabyte, multi-language monorepo — used by more than 2,000 engineers — onto CloudBuild and the Office-wide NMake-to-MSBuild platform, replacing a legacy pipeline that had no caching or distributed execution.
- Owned SharePoint's dependency-graph reconstruction and correctness validation for the migration, enabling affected-target incremental builds by identifying modified components and their transitive downstream dependents, and cutting incremental build time by about 10x while preserving the full-build correctness baseline.
- Designed and implemented binary-equivalence and deterministic-build diagnostics that isolated environmental noise such as timestamps and file paths — the same class of nondeterminism that silently breaks cross-machine cache reuse — and verified migrated MSBuild outputs against legacy NMake artifacts.
- Collaborated with the upstream Office build-platform team and SharePoint feature teams to integrate platform changes without breaking build correctness or daily workflows for more than 2,000 downstream engineers.

SELECTED PROJECT

MyPawStories | Creator & Lead Engineer

2026 – Present

- Built end-to-end CI/CD and release pipelines with GitHub Actions, Terraform, Docker, and GCP, automating 1,500+ tests, QA promotion, smoke validation, and controlled production releases; added Bazel as the unified build and test entry point, including custom build rules.
- Owned the product end to end across three React/TypeScript applications, two .NET services, PostgreSQL, asynchronous AI generation, Stripe payment flows, and production operations.
- Built an AI-assisted customer-support agent as a Codex plugin with a scoped MCP server and 26 task-specific tools, enabling partners to resolve production cases through least-privilege, approval-gated, and auditable workflows.

EDUCATION

South China University of Technology

Guangzhou, China

Master of Science, Computer Application Technology

2012 – 2015

Bachelor of Science, Computer Science

2008 – 2012